

CloudWatch Agent Setup on EC2 (Complete Guide)

◆ 1. Introduction

Amazon CloudWatch Agent is a tool used to:

- Collect system metrics (CPU, Memory, Disk)
 - Collect logs (application logs, PM2 logs)
 - Send data to CloudWatch for monitoring
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◆ 2. Prerequisites

Before installation, ensure:

- EC2 instance is running (**Amazon EC2**)
- Internet access available
- IAM Role attached with policy:

👉 `CloudWatchAgentServerPolicy` ✓

◆ 3. Install CloudWatch Agent (Ubuntu)

Step 1: Download latest package

wget

`https://s3.amazonaws.com/amazoncloudwatch-agent/ubuntu/amd64/latest/amazon-cloudwatch-agent.deb`

Step 2: Install package

`sudo dpkg -i amazon-cloudwatch-agent.deb`

Step 3: Fix dependencies

```
sudo apt-get install -f -y
```

Step 4: Verify installation

```
/opt/aws/amazon-cloudwatch-agent/bin/amazon-cloudwatch-agent-ctl -h
```

◆ 4. Create Configuration File

Run wizard:

```
sudo /opt/aws/amazon-cloudwatch-agent/bin/amazon-cloudwatch-agent-config-wizard
```

👉 It will ask questions and generate config file

Default path:

```
/opt/aws/amazon-cloudwatch-agent/bin/config.json
```

◆ 5. Example Config (Metrics + Logs)

```
{
  "agent": {
    "metrics_collection_interval": 60,
    "run_as_user": "cwagent"
  },
  "logs": {
    "logs_collected": {
      "files": {
        "collect_list": [
          {
            "file_path": "/home/ubuntu/.pm2/logs/portfolio-dev-out.log",
            "log_group_name": "pm2-out",
            "log_stream_name": "{instance_id}",
            "retention_in_days": 7
          },
          {
            "file_path": "/home/ubuntu/.pm2/logs/portfolio-dev-error.log",
            "log_group_name": "pm2-error",
            "log_stream_name": "{instance_id}",
            "retention_in_days": 7
          }
        ]
      }
    }
  }
}
```

```

    }
  ]
}
},
"metrics": {
  "append_dimensions": {
    "InstanceId": "${aws:InstanceId}"
  },
  "metrics_collected": {
    "cpu": {
      "measurement": ["cpu_usage_idle", "cpu_usage_user"],
      "metrics_collection_interval": 60
    },
    "mem": {
      "measurement": ["mem_used_percent"]
    },
    "disk": {
      "measurement": ["used_percent"],
      "resources": ["*"]
    }
  }
}
}
}

```

◆ 6. Start CloudWatch Agent

```

sudo /opt/aws/amazon-cloudwatch-agent/bin/amazon-cloudwatch-agent-ctl \
-a fetch-config \
-m ec2 \
-c file:/opt/aws/amazon-cloudwatch-agent/bin/config.json \
-s

```

◆ 7. Check Status

```

sudo /opt/aws/amazon-cloudwatch-agent/bin/amazon-cloudwatch-agent-ctl -m ec2 -a status

```

👉 Output:

```

{
  "status": "running"
}

```

♦ 8. Verify Logs & Metrics

👉 Go to AWS Console:

- CloudWatch → Logs → Log groups
- CloudWatch → Metrics

♦ 9. Common Issues & Fixes

✗ Issue 1: Logs not showing

👉 Fix:

- Use absolute file path
- Check IAM role

✗ Issue 2: Metrics not showing

👉 Fix:

- Attach `CloudWatchAgentServerPolicy`
- Restart agent

✗ Issue 3: collectd error

```
sudo apt install collectd -y
```

♦ 10. Restart Agent

```
sudo systemctl restart amazon-cloudwatch-agent
```

🔥 Final Summary

👉 Steps:

1. Install agent
 2. Create config.json
 3. Attach IAM role
 4. Start agent
 5. Verify logs & metrics
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One-line Conclusion



CloudWatch Agent is used to collect system metrics and logs from EC2 and send them to CloudWatch for monitoring and analysis.